
Chapter 13 Principles of Bioenergetics

Multiple Choice Questions

1. Bioenergetics and thermodynamics

Page: 506 Difficulty: 1 Ans: E

If the ΔG° of the reaction $A \rightarrow B$ is -40 kJ/mol , under standard conditions the reaction:

- A) is at equilibrium.
- B) will never reach equilibrium.
- C) will not occur spontaneously.
- D) will proceed at a rapid rate.
- E) will proceed spontaneously from A to B.

2. Bioenergetics and thermodynamics

Page: 506 Difficulty: 1 Ans: C

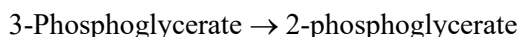
For the reaction $A \rightarrow B$, $\Delta G^\circ = -60 \text{ kJ/mol}$. The reaction is started with 10 mmol of A; no B is initially present. After 24 hours, analysis reveals the presence of 2 mmol of B, 8 mmol of A. Which is the *most* likely explanation?

- A) A and B have reached equilibrium concentrations.
- B) An enzyme has shifted the equilibrium toward A.
- C) B formation is kinetically slow; equilibrium has not been reached by 24 hours.
- D) Formation of B is thermodynamically unfavorable.
- E) The result described is impossible, given the fact that ΔG° is -60 kJ/mol .

3. Bioenergetics and thermodynamics

Page: 506 Difficulty: 2 Ans: A

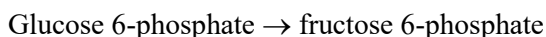
When a mixture of 3-phosphoglycerate and 2-phosphoglycerate is incubated at 25°C with phosphoglycerate mutase until equilibrium is reached, the final mixture contains six times as much 2-phosphoglycerate as 3-phosphoglycerate. Which one of the following statements is *most* nearly correct, when applied to the reaction as written? ($R = 8.315 \text{ J/mol}\cdot\text{K}$; $T = 298 \text{ K}$)



- A) ΔG° is -4.44 kJ/mol .
- B) ΔG° is zero.
- C) ΔG° is $+12.7 \text{ kJ/mol}$.
- D) ΔG° is incalculably large and positive.
- E) ΔG° cannot be calculated from the information given.

4. Bioenergetics and thermodynamics**Page: 506 Difficulty: 2 Ans: D**

When a mixture of glucose 6-phosphate and fructose 6-phosphate is incubated with the enzyme phosphohexose isomerase (which catalyzes the interconversion of these two compounds) until equilibrium is reached, the final mixture contains twice as much glucose 6-phosphate as fructose 6-phosphate. Which one of the following statements is *best* applied to this reaction outlined below? ($R = 8.315 \text{ J/mol}\cdot\text{K}$; $T = 298 \text{ K}$)



- A) ΔG° is incalculably large and negative.
- B) ΔG° is -1.72 kJ/mol .
- C) ΔG° is zero.
- D) ΔG° is $+1.72 \text{ kJ/mol}$.
- E) ΔG° is incalculably large and positive.

5. Bioenergetics and thermodynamics**Page: 506 Difficulty: 1 Ans: A**

Hydrolysis of 1 M glucose 6-phosphate catalyzed by glucose 6-phosphatase is 99% complete at equilibrium (i.e., only 1% of the substrate remains). Which of the following statements is *most* nearly correct? ($R = 8.315 \text{ J/mol}\cdot\text{K}$; $T = 298 \text{ K}$)

- A) ΔG° is -11 kJ/mol .
- B) ΔG° is -5 kJ/mol .
- C) ΔG° is 0 kJ/mol .
- D) ΔG° is $+11 \text{ kJ/mol}$.
- E) ΔG° cannot be determined from the information given.

6. Bioenergetics and thermodynamics**Page: 506 Difficulty: 2 Ans: C**

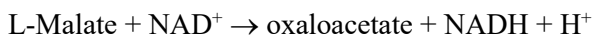
The reaction $A + B \rightarrow C$ has a ΔG° of -20 kJ/mol at 25°C . Starting under standard conditions, one can predict that:

- A) at equilibrium, the concentration of B will exceed the concentration of A.
- B) at equilibrium, the concentration of C will be less than the concentration of A.
- C) at equilibrium, the concentration of C will be much greater than the concentration of A or B.
- D) C will rapidly break down to A + B.
- E) when A and B are mixed, the reaction will proceed rapidly toward formation of C.

7. Bioenergetics and thermodynamics**Page: 509 Difficulty: 1 Ans: A**

Which of the following compounds has the largest negative value for the standard free-energy change (ΔG°) upon hydrolysis?

- A) Acetic anhydride
- B) Glucose 6-phosphate
- C) Glutamine
- D) Glycerol 3-phosphate
- E) Lactose

8. Bioenergetics and thermodynamics**Page: 509 Difficulty: 1 Ans: E**For the following reaction, $\Delta G^\circ = +29.7$ kJ/mol.

The reaction as written:

- A) can never occur in a cell.
- B) can occur in a cell only if it is coupled to another reaction for which ΔG° is positive.
- C) can occur only in a cell in which NADH is converted to NAD^+ by electron transport.
- D) cannot occur because of its large activation energy.
- E) may occur in cells at some concentrations of substrate and product.

9. Bioenergetics and thermodynamics**Page: 508 Difficulty: 1 Ans: A**For the reaction $\text{A} \rightarrow \text{B}$, the K_{eq} is 10^4 . If a reaction mixture originally contains 1 mmol of A and no B, which one of the following must be *true*?

- A) At equilibrium, there will be far more B than A.
- B) The rate of the reaction is very slow.
- C) The reaction requires coupling to an exergonic reaction in order to proceed.
- D) The reaction will proceed toward B at a very high rate.
- E) ΔG° for the reaction will be large and positive.

10. Bioenergetics and thermodynamics**Page: 508 Difficulty: 1 Ans: C, E**For the reaction $\text{A} \rightarrow \text{B}$, the K_{eq} is 10^{-6} . If a reaction mixture originally contains 1 mmol of A and 1 mmol of B, which one of the following must be *true*?

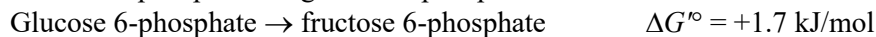
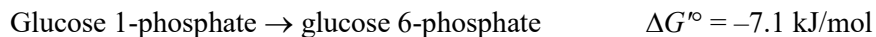
- A) At equilibrium, there will be still be equal levels of A and B.
- B) The rate of the reaction is very slow.
- C) At equilibrium, the amount of A will greatly exceed the amount of B.
- D) The reaction will proceed toward B at a very high rate.
- E) ΔG° for the reaction will be large and positive.

11. Bioenergetics and thermodynamics**Page: 509 Difficulty: 2 Ans: A**In glycolysis, fructose 1,6-bisphosphate is converted to two products with a standard free-energy change (ΔG°) of 23.8 kJ/mol. Under what conditions encountered in a normal cell will the free-energy change (ΔG) be negative, enabling the reaction to proceed spontaneously to the right?

- A) Under standard conditions, enough energy is released to drive the reaction to the right.
- B) The reaction will not go to the right spontaneously under any conditions because the ΔG° is positive.
- C) The reaction will proceed spontaneously to the right if there is a high concentration of products relative to the concentration of fructose 1,6-bisphosphate.
- D) The reaction will proceed spontaneously to the right if there is a high concentration of fructose 1,6-bisphosphate relative to the concentration of products.
- E) None of the above conditions is sufficient.

12. Bioenergetics and thermodynamics**Pages: 510–511 Difficulty: 2 Ans: C**

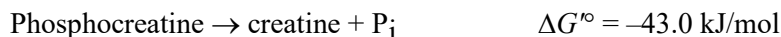
During glycolysis, glucose 1-phosphate is converted to fructose 6-phosphate in two successive reactions:

 $\Delta G'^{\circ}$ for the overall reaction is:

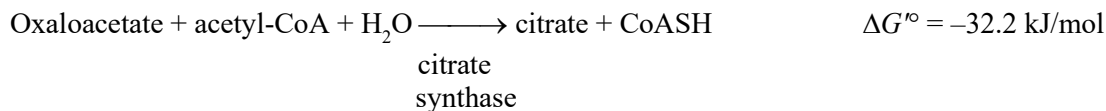
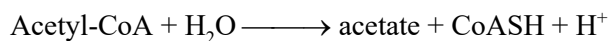
- A) -8.8 kJ/mol.
- B) -7.1 kJ/mol.
- C) -5.4 kJ/mol.
- D) +5.4 kJ/mol.
- E) +8.8 kJ/mol.

13. Bioenergetics and thermodynamics**Pages: 510–511 Difficulty: 2 Ans: B**

The standard free-energy changes for the reactions below are given.

What is the overall $\Delta G'^{\circ}$ for the following reaction?

- A) -73.5 kJ/mol
- B) -12.5 kJ/mol
- C) +12.5 kJ/mol
- D) +73.5 kJ/mol
- E) $\Delta G'^{\circ}$ cannot be calculated without K_{eq}' .

14. Bioenergetics and thermodynamics**Pages: 510–511 Difficulty: 2 Ans: C**The $\Delta G'^{\circ}$ values for the two reactions shown below are given.What is the $\Delta G'^{\circ}$ for the hydrolysis of acetyl-CoA?

- A) -34.1 kJ/mol
- B) -32.2 kJ/mol
- C) -30.3 kJ/mol
- D) +61.9 kJ/mol
- E) +34.1 kJ/mol

15. Chemical Logic and Common Biochemical Reactions**Page: 512 Difficulty: 1 Ans: A**Which of the following is *not* nucleophilic?

- A) A proton
- B) A carbanion
- C) An imidazole
- D) A hydroxide
- E) A carboxylic acid

16. Chemical Logic and Common Biochemical Reactions**Page: 512 Difficulty: 1 Ans: B**Which of the following is *not* electrophilic?

- A) A proton
- B) A sulfhydryl
- C) A protonated imine
- D) A carbonyl group
- E) A phosphoryl group

17. Chemical Logic and Common Biochemical Reactions**Page: 513 Difficulty: 2 Ans: D**Which of the following is *not* true?

- A) The carbon adjacent to a carbonyl can be resonance stabilized to form a carbanion.
- B) A carbonyl carbon can be made more electrophilic by a nearby metal ion.
- C) The carbon adjacent to an imine can be resonance stabilized to form a carbanion
- D) Decarboxylation of an α -keto acid goes through a carbocation intermediate.
- E) A Claisen ester condensation reaction goes through a carbanion intermediate.

18. Chemical Logic and Common Biochemical Reactions**Pages: 515–516 Difficulty: 2 Ans: D**The reaction $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ is an example of a _____ reaction.

- A) homolytic cleavage
- B) internal rearrangement
- C) free radical
- D) group transfer
- E) oxidation/reduction

19. Chemical Logic and Common Biochemical Reactions**Page: 516 Difficulty: 2 Ans: E**Which of the following is *true* about oxidation-reduction reactions?

- A) They usually proceed through homolytic cleavage.

- B) During oxidation a compound gains electrons.
- C) Dehydrogenases typically remove two electrons and two hydrides.
- D) There are four commonly accessed oxidation states of carbon.
- E) Every oxidation must be accompanied by a reduction.

20. Phosphoryl group transfers and ATP**Page: 518 Difficulty: 2 Ans: B**

All of the following contribute to the large, negative, free-energy change upon hydrolysis of “high-energy” compounds *except*:

- A) electrostatic repulsion in the reactant.
- B) low activation energy of forward reaction.
- C) stabilization of products by extra resonance forms.
- D) stabilization of products by ionization.
- E) stabilization of products by solvation.

21. Phosphoryl group transfers and ATP**Page: 519 Difficulty: 1 Ans: E**

The hydrolysis of ATP has a large negative $\Delta G'^{\circ}$; nevertheless it is stable in solution due to:

- A) entropy stabilization.
- B) ionization of the phosphates.
- C) resonance stabilization.
- D) the hydrolysis reaction being endergonic.
- E) the hydrolysis reaction having a large activation energy.

22. Phosphoryl group transfers and ATP**Page: 520 Difficulty: 3 Ans: E**

The hydrolysis of phosphoenolpyruvate proceeds with a $\Delta G'^{\circ}$ of about -62 kJ/mol. The greatest contributing factors to this reaction are the destabilization of the reactants by electrostatic repulsion and stabilization of the product pyruvate by:

- A) electrostatic attraction.
- B) ionization.
- C) polarization.
- D) resonance.
- E) tautomerization.

23. Phosphoryl group transfers and ATP**Page: 520 Difficulty: 1 Ans: B**

Which one of the following compounds does *not* have a large negative free energy of hydrolysis?

- A) 1,3-bis phosphoglycerate
- B) 3-phosphoglycerate
- C) ADP
- D) Phosphoenolpyruvate
- E) Thioesters (e.g. acetyl-CoA)

24. Phosphoryl group transfers and ATP**Page: 524 Difficulty: 1 Ans: B**

The immediate precursors of DNA and RNA synthesis in the cell all contain:

- A) 3' triphosphates.
- B) 5' triphosphates.
- C) adenine.
- D) deoxyribose.
- E) ribose.

25. Phosphoryl group transfers and ATP**Pages: 525-526 Difficulty: 2 Ans: A**

Muscle contraction involves the conversion of:

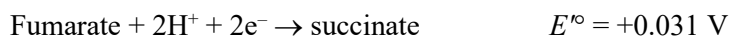
- A) chemical energy to kinetic energy.
- B) chemical energy to potential energy.
- C) kinetic energy to chemical energy.
- D) potential energy to chemical energy.
- E) potential energy to kinetic energy.

26. Biological oxidation-reduction reactions**Page: 528 Difficulty: 1 Ans: D**Biological oxidation-reduction reactions *always* involve:

- A) direct participation of oxygen.
- B) formation of water.
- C) mitochondria.
- D) transfer of electron(s).
- E) transfer of hydrogens.

27. Biological oxidation-reduction reactions**Page: 528 Difficulty: 2 Ans: B**Biological oxidation-reduction reactions *never* involve:

- A) transfer of e^- from one molecule to another.
- B) formation of free e^- .
- C) transfer of H^+ (or H_3O^+) from one molecule to another.
- D) formation of free H^+ (or H_3O^+).
- E) none of the above.

28. Biological oxidation-reduction reactions**Pages: 530-531 Difficulty: 2 Ans: B**The standard reduction potentials (E°) for the following half reactions are given.

If you mixed succinate, fumarate, FAD, and FADH₂ together, all at 1 M concentrations and in the presence of succinate dehydrogenase, which of the following would happen *initially*?

- A) Fumarate and succinate would become oxidized; FAD and FADH₂ would become reduced.
- B) Fumarate would become reduced; FADH₂ would become oxidized.
- C) No reaction would occur because all reactants and products are already at their standard

concentrations.

- D) Succinate would become oxidized; FAD would become reduced.
- E) Succinate would become oxidized; FADH_2 would be unchanged because it is a cofactor.

29. Biological oxidation-reduction reactions

Pages: 530–531 Difficulty: 3 Ans: E

E° of the NAD^+/NADH half reaction is -0.32 V . The E° of the oxaloacetate/malate half reaction is -0.175 V . When the concentrations of NAD^+ , NADH , oxaloacetate, and malate are all 10^{-5} M , the “spontaneous” reaction is:

- A) $\text{malate} + \text{NAD}^+ \rightarrow \text{oxaloacetate} + \text{NADH} + \text{H}^+$.
- B) $\text{malate} + \text{NADH} + \text{H}^+ \rightarrow \text{oxaloacetate} + \text{NAD}^+$.
- C) $\text{NAD}^+ + \text{NADH} + \text{H}^+ \rightarrow \text{malate} + \text{oxaloacetate}$.
- D) $\text{NAD}^+ + \text{oxaloacetate} \rightarrow \text{NADH} + \text{H}^+ + \text{malate}$.
- E) $\text{oxaloacetate} + \text{NADH} + \text{H}^+ \rightarrow \text{malate} + \text{NAD}^+$.

30. Biological oxidation-reduction reactions

Pages: 532–533 Difficulty: 2 Ans: A

The structure of NAD^+ does *not* include:

- A) a flavin nucleotide.
- B) a pyrophosphate bond.
- C) an adenine nucleotide.
- D) nicotinamide.
- E) two ribose residues.

31. Biological oxidation-reduction reactions

Pages: 532–533 Difficulty: 2 Ans: C

Which of the following is *not* true for the nicotinamide cofactors?

- A) The oxidized form is positively charged.
- B) The reduced form has a large extinction coefficient at 340 nm .
- C) The oxidized form provides reducing equivalents to other molecules.
- D) Oxidation-reduction reactions with nicotinamides usually involve hydride transfer.
- E) Enzymes transfer hydrides stereospecifically to one or the other side of the nicotinamide ring.

Short Answer Questions

32. Bioenergetics and thermodynamics

Pages: 506–507 Difficulty: 2

Explain the relationships among the change in the degree of order, the change in entropy, and the change in free energy that occur during a chemical reaction.

Ans: Entropy is a measure of disorder. Thus, if there is an increase in order there is a decrease in entropy. The greater the entropy of a system, the smaller its free energy. Thus, an increase in entropy during a reaction will result in a decrease in free energy.

33. Bioenergetics and thermodynamics

Pages: 507–508 Difficulty: 2

Consider the reaction: $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$. If the equilibrium constant for this reaction is a large number

(say, 10,000), what do we know about the standard free-energy change (ΔG°) for the reaction? Describe the relationship between K_{eq}' and ΔG° .

Ans: $\Delta G^\circ = -RT \ln K_{eq}'$. If K_{eq}' is a large (positive) number, the term $-RT \ln K_{eq}'$ (and therefore ΔG°) has a relatively large, negative value.

34. Bioenergetics and thermodynamics

Page: 508 Difficulty: 2

The standard free energy change (ΔG°) for ATP hydrolysis is -30.5 kJ/mol. ATP, ADP, and P_i are mixed together at initial concentrations of 1 M of each, then left alone until the reaction $ADP + P_i \rightarrow ATP$ has come to equilibrium. For each species (i.e., ATP, ADP, and P_i), indicate whether the concentration will be equal to 1 M, less than 1 M, or greater than 1 M.

Ans: At equilibrium, $ATP < 1$ M; $ADP > 1$ M; $P_i > 1$ M.

35. Bioenergetics and thermodynamics

Page: 508 Difficulty: 3

If a 0.1 M solution of glucose 1-phosphate is incubated with a catalytic amount of phospho-glucomutase, the glucose 1-phosphate is transformed to glucose 6-phosphate until equilibrium is reached. At equilibrium, the concentration of glucose 1-phosphate is 4.5×10^{-3} M and that of glucose 6-phosphate is 8.6×10^{-2} M. Set up the expressions for the calculation of K_{eq}' and ΔG° for this reaction (in the direction of glucose 6-phosphate formation). ($R = 8.315$ J/mol·K; $T = 298$ K)

Ans: $K_{eq}' = \frac{[\text{glucose 6-phosphate}]}{[\text{glucose 1-phosphate}]} = \frac{0.086 \text{ M}}{0.0045 \text{ M}} = 19$

$$\Delta G^\circ = -RT \ln K_{eq}' = -(8.315 \text{ J/mol}\cdot\text{K})(298 \text{ K})(\ln 19) = -7.3 \text{ kJ/mol}$$

36. Bioenergetics and thermodynamics

Page: 508 Difficulty: 2

What is the difference between ΔG and ΔG° of a chemical reaction? Describe, quantitatively, the relationship between them.

Ans: ΔG° is the difference in free energies of the products and reactants at standard conditions. Thus, it is a physical constant, characteristic of each chemical reaction. ΔG is a variable that depends on ΔG° , the temperature, and the concentrations of all reactants and products:

$$\Delta G = \Delta G^\circ + RT \ln \frac{[\text{product}]}{[\text{reactant}]}$$

37. Bioenergetics and thermodynamics

Page: 508 Difficulty: 2

The expression $\Delta G = \Delta G^\circ + RT \ln K_{eq}'$ for the actual free-energy change for the reaction $A + B \rightarrow C + D$ is incorrect. Why is it wrong, and what is the correct expression for the real free-energy change of this reaction?

Ans: The correct expression is: $\Delta G = \Delta G^\circ + RT \ln \frac{[C][D]}{[A][B]}$

Substitution of K_{eq}' for the concentration term is valid only for the special case in which these

concentrations represent the *equilibrium* concentrations. In this situation, $\Delta G = 0$.

38. Bioenergetics and thermodynamics

Page: 509 Difficulty: 3

Explain in quantitative terms the circumstances under which the following reaction can proceed.



Ans: A reaction for which ΔG° is positive can proceed under conditions in which ΔG is negative. From the relationship

$$\Delta G = \Delta G^{\circ} + RT \ln \frac{[\text{product}]}{[\text{reactant}]}$$

it is clear that if the concentration of product is kept very low (by its subsequent metabolic removal, for instance), the logarithmic term becomes negative and ΔG can then have a negative value.

39. Bioenergetics and thermodynamics

Pages: 506–509 Difficulty: 2

Explain why each of the following statements is *false*.

- (a) In a reaction under standard conditions, only the reactants are fixed at 1 M.
- (b) When ΔG° is positive, $K_{\text{eq}}' > 1$.
- (c) ΔG and ΔG° mean the same thing.
- (d) When $\Delta G^{\circ} = 1.0 \text{ kJ/mol}$, $K_{\text{eq}}' = 1$.

Ans: (a) Both reactants and products are fixed at 1 M. (b) $\Delta G^{\circ} = -RT \ln K_{\text{eq}}'$, so when $\Delta G^{\circ} < 0$, the term $-RT \ln K_{\text{eq}}'$ has a negative value and $K_{\text{eq}}' > 1$. (c) ΔG° is the difference in free energies of the products and reactants at standard conditions, and is a constant, characteristic of each chemical reaction. ΔG is a variable that depends on ΔG° , the temperature, and the concentrations of all reactants and products

$$\Delta G = \Delta G^{\circ} + RT \ln \frac{[\text{product}]}{[\text{reactant}]}$$

(d) $\Delta G^{\circ} = -RT \ln K_{\text{eq}}'$. When $\Delta G^{\circ} = 1$, $K_{\text{eq}}' < 1$.

40. Bioenergetics and thermodynamics

Pages: 408–511 Difficulty: 3

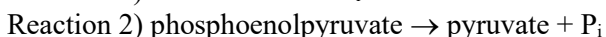
In glycolysis, the enzyme pyruvate kinase catalyzes this reaction:



Given the information below, show how you would calculate the equilibrium constant for this reaction. ($R = 8.315 \text{ J/mol} \cdot \text{K}$; $T = 298 \text{ K}$)



$$\Delta G^{\circ} = -30.5 \text{ kJ/mol}$$



$$\Delta G^{\circ} = -61.9 \text{ kJ/mol}$$

Ans: The reaction is the sum of reaction 2 and the reverse of reaction 1. Therefore, $\Delta G^{\circ} = -31.4 \text{ kJ/mol}$.

$$\begin{aligned}\Delta G'^{\circ} &= -RT \ln K_{\text{eq}}' \\ \ln K_{\text{eq}}' &= -\Delta G'^{\circ}/RT \\ &= 31.4 \text{ kJ/mol} / [(8.315 \text{ J/mol}\cdot\text{K})(298 \text{ K})] \\ \ln K_{\text{eq}}' &= 12.672 \\ K_{\text{eq}}' &= 3.19 \times 10^5\end{aligned}$$

41. Bioenergetics and thermodynamics**Page: 510 Difficulty: 2**

Explain what is meant by the statement: “Standard free-energy changes are additive.” Give an example of the usefulness of this additive property in understanding how cells carry out thermodynamically unfavorable chemical reactions.

Ans: When two chemical reactions can be summed to give a third reaction, the standard free-energy change for the third reaction is the arithmetic sum of the standard free-energy changes of the other two reactions. Many possible examples may be given. (See p. 19 for one such example.)

42. Bioenergetics and thermodynamics**Pages: 508–511 Difficulty: 2**

Given $\Delta G'^{\circ}$ for each of the following reactions,

- | | |
|---|--|
| 1. $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ | $\Delta G'^{\circ} = -30.5 \text{ kJ/mol}$ |
| 2. $\text{glucose 6-phosphate} \rightarrow \text{glucose} + \text{P}_i$ | $\Delta G'^{\circ} = -13.8 \text{ kJ/mol}$ |

show how you would calculate the standard free-energy change ($\Delta G'^{\circ}$) for the reaction:

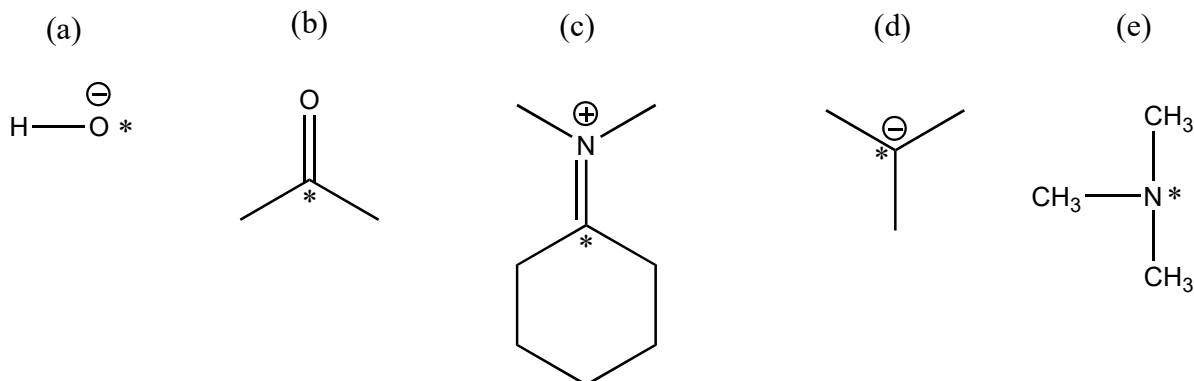
3. $\text{ATP} + \text{glucose} \rightarrow \text{glucose 6-phosphate} + \text{ADP}$

Ans: Reaction 3 is the sum of reaction 1 and the reversal of reaction 2. Because of the additivity of free energy changes, the overall $\Delta G'^{\circ}$ for reaction 3 is the sum of the free energy changes for reaction 1 and the reversal of reaction 2:

- | | |
|--|--|
| 1. $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ | $\Delta G_1'^{\circ} = -30.5 \text{ kJ/mol}$ |
| 2. $\text{glucose} + \text{P}_i \rightarrow \text{glucose 6-phosphate}$ | $\Delta G_2'^{\circ} = +13.8 \text{ kJ/mol}$ |
| 3. $\text{ATP} + \text{glucose} \rightarrow \text{ADP} + \text{glucose 6-phosphate}$ | |
| $\Delta G_3'^{\circ} = \Delta G_1'^{\circ} + \Delta G_2'^{\circ} = (-30.5 + 13.8) \text{ kJ/mol} = -16.7 \text{ kJ/mol}$ | |

43. Chemical Logic and Common Biochemical Reactions**Page: 512 Difficulty: 2**

Classify each of the *ed atoms as an electrophile or a nucleophile:



Ans:

- (a) Nucleophile
- (b) Electrophile
- (c) Electrophile
- (d) Nucleophile
- (e) Nucleophile

44. Phosphoryl group transfers and ATP

Pages: 517–519 Difficulty: 2

Why is the actual free energy (ΔG) of hydrolysis of ATP in the cell different from the standard free energy (ΔG°)?

Ans: The concentrations of the reactant (ATP) and the products (ADP and P_i) are not all equal to 1 M. The pH is not exactly equal to 7.0 (the standard pH) and the temperature is usually higher than the standard temperature of 25 °C.

45. Phosphoryl group transfers and ATP

Page: 520 Difficulty: 2

The free energy of hydrolysis of phosphoenolpyruvate is -61.9 kJ/mol. Rationalize this large, negative value for ΔG° in chemical terms.

Ans: The product of hydrolysis is enol pyruvate, which quickly tautomerizes to pyruvate. This keto-enol tautomerization stabilizes the products of hydrolysis relative to phosphoenolpyruvate, which cannot tautomerize.

46. Phosphoryl group transfers and ATP

Page: 522 Difficulty: 3

In general, when ATP hydrolysis is coupled to an energy-requiring reaction, the actual reaction often consists of the transfer of a phosphate group from ATP to another substrate, rather than an actual hydrolysis of the ATP. Explain.

Ans: Hydrolysis of the ATP would result in the loss of most of the free energy as heat. In a transfer reaction, the gamma (third) phosphate of ATP is transferred to the reaction substrate to produce a high-energy phosphorylated intermediate, which can then form the product in an exergonic reaction.

47. Phosphoryl group transfers and ATP

Pages: 524–526 Difficulty: 2

The first law of thermodynamics states that the amount of energy in the universe is constant, but that the various forms of energy can be interconverted. Describe four different types of such energy

transduction that occur in living organisms and provide one example for each.

Ans: Conversion of:

- chemical to osmotic energy (ATP-driven active transport across a membrane)
- electromagnetic to electrical energy (light-induced electron flow in chloroplasts)
- chemical to electromagnetic energy (ATP-dependent photon emission in fireflies)
- chemical to mechanical energy (ATP-driven muscle contraction)

Other answers are possible.

48. Biological oxidation-reduction reactions

Page: 528 Difficulty: 1

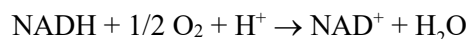
What is an oxidation? What is a reduction? Can an oxidation occur without a simultaneous reduction? Why or why not?

Ans: Oxidation is the loss of electrons; reduction is the gain of electrons. Free electrons are unstable (do not occur), so whenever an electron is released by oxidation of some species, an electron must be accepted by reduction of another species.

49. Biological oxidation-reduction reactions

Page: 531 Difficulty: 3

During transfer of two electrons through the mitochondrial respiratory chain, the overall reaction is:



For this reaction, the difference in reduction potentials for the two half-reactions (ΔE°) is +1.14 V. Show how you would calculate the standard free-energy change, ΔG° , for the reaction. (The Faraday constant, $\mathfrak{F}$, is 96.48 kJ/V·mol.)

Ans: $\Delta G^\circ = -n \mathfrak{F} \Delta E^\circ$

For reactions involving NADH, two electrons are transferred ($n = 2$). So $\Delta G^\circ = (-2)(96.48 \text{ kJ/V} \cdot \text{mol})(1.14 \text{ V}) = -220 \text{ kJ/mol}$.

50. Biological oxidation-reduction reactions

Page: 531 Difficulty: 2

If ΔE° for an oxidation-reduction reaction is positive, will ΔG° be positive or negative? What is the equation that relates ΔG° and ΔE° ?

Ans: Negative. $\Delta G^\circ = -n \mathfrak{F} \Delta E^\circ$.

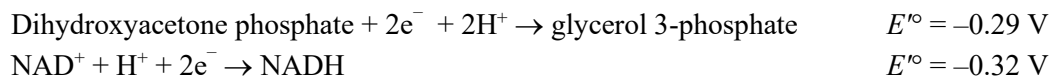
51. Biological oxidation-reduction reactions

Pages: 531–532 Difficulty: 2

Glycerol 3-phosphate dehydrogenase catalyzes the following reversible reaction:



Given the standard reduction potentials below, calculate ΔG° for the glycerol 3-phosphate dehydrogenase reaction, proceeding from left to right as shown. Show your work. (The Faraday constant, $\mathfrak{F}$, is 96.48 kJ/V·mol.)



Ans:

$$\begin{aligned} \Delta E^\circ &= E^\circ (\text{electron acceptor}) - E^\circ (\text{electron donor}) \\ &= -0.32 - (-0.29) = -0.03 \text{ V} \\ \Delta G^\circ &= -n \mathfrak{F} \Delta E^\circ = (-2)(96.48 \text{ kJ/V} \cdot \text{mol})(-0.03 \text{ V}) \\ \Delta G^\circ &= +5.8 \text{ kJ/mol} \end{aligned}$$

52. Biological oxidation-reduction reactions

Page: 531 Difficulty: 1

For each pair of ions or compounds below, indicate which is the more highly reduced species.

- (a) $\text{Co}^{2+}/\text{Co}^+$
- (b) Glucose/ CO_2
- (c) $\text{Fe}^{3+}/\text{Fe}^{2+}$
- (d) Acetate/ CO_2
- (e) Ethanol/acetic acid
- (f) Acetic acid/acetaldehyde

Ans: (a) Co^+ ; (b) glucose; (c) Fe^{2+} ; (d) acetate; (e) ethanol; (f) acetaldehyde

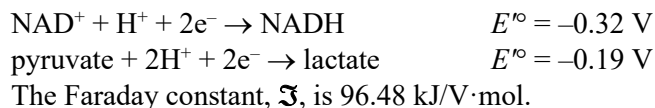
53. Biological oxidation-reduction reactions

Pages: 531–532 Difficulty: 3

Lactate dehydrogenase catalyzes the reversible reaction:



Given the following facts (a) tell in which direction the reaction will tend to go if NAD^+ , NADH , pyruvate, and lactate were mixed, all at 1 M concentrations, in the presence of lactate dehydrogenase at pH 7; (b) calculate ΔG° for this reaction. Show your work.

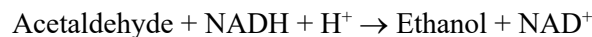


Ans: $\Delta G^\circ = E^\circ (\text{acceptor}) - E^\circ (\text{donor}) = -0.19 \text{ V} - (-0.32 \text{ V}) = +0.13 \text{ V}$
 $\Delta G^\circ = -n \mathfrak{F} \Delta E^\circ = (-2)(96.48 \text{ kJ/V} \cdot \text{mol})(0.13 \text{ V})$
 $\Delta G^\circ = -25.1 \text{ kJ/mol}$

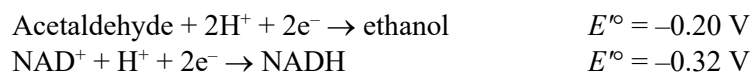
54. Biological oxidation-reduction reactions

Pages: 531–532 Difficulty: 3

Alcohol dehydrogenase catalyzes the following reversible reaction:



Use the following information to answer the questions below:



The Faraday constant, $\mathfrak{F}$, is $96.48 \text{ kJ/V} \cdot \text{mol}$.

- Calculate ΔG° for the reaction as written. Show your work.
- Given your answer to (a), what is the ΔG° for the reaction occurring in the reverse direction?
- Which reaction (forward or reverse) will tend to occur spontaneously under standard conditions?
- In the cell, the reaction actually proceeds in the direction that has a positive ΔG° . Explain how this could be possible.

Ans:

$$\begin{aligned} \text{(a)} \quad \Delta G^\circ &= E^\circ(\text{acceptor}) - E^\circ(\text{donor}) = -0.20 - (-0.32) = +0.12 \text{ V} \\ \Delta G^\circ &= -n\mathfrak{F}\Delta E^\circ = (-2)(96.48 \text{ kJ/V} \cdot \text{mol})(0.12 \text{ V}) = -23.2 \text{ kJ/mol} \end{aligned}$$

- For the reverse reaction, $\Delta G^\circ = +23.2 \text{ kJ/mol}$.
 - The forward reaction will occur spontaneously.
 - A reaction for which ΔG° is positive can occur if ΔG is negative.
- From the relationship

$$\Delta G = \Delta G^\circ + RT \ln \frac{[\text{product}]}{[\text{reactant}]}$$

it is clear that if the concentration of product is kept very low (by its removal in a subsequent metabolic step), the log term becomes negative and ΔG can have a negative value.